

AI AGENT BOOTCAMP

Day 1 Handout

The Agentic Mindset

This handout summarises the core concepts from Day 1. By the end of today you will understand the structural difference between a standard LLM and a true AI Agent, and how autonomous reasoning loops power agents to solve complex, real-world tasks.

1. LLMs vs. Agents: From Knowledge to Action

The Librarian vs. the Executive Assistant

- **LLM:** The LLM is the Librarian.

Has read every book in the building and can give brilliant answers — but stays behind the desk. It cannot take real-world actions.

- **Agent:** The Agent is the Executive Assistant.

Uses the librarian's knowledge AND has a phone, a computer, and a credit card. It can leave the library, use tools, and complete a multi-step mission.

Key Differences at a Glance

Feature	LLM (The Brain)	AI Agent (The Body + Brain)
Input	A prompt	A high-level goal
Output	Text / Code	A completed task or result
Tools	None (internal knowledge only)	Browsers, APIs, databases, etc.
Autonomy	Low — needs step-by-step instructions	High — figures out the 'how' on its own

In short: an LLM predicts the next word; an Agent uses that predictive power to decide which action to take next.

2. Autonomous Loops: The Heartbeat of an Agent

Agents think using an Autonomous Loop — most commonly the ReAct (Reason + Act) framework. Instead of generating a final answer immediately, the agent cycles through:

1. Observe — look at the current state or the user's request.
2. Think — reason about what needs to happen next.
3. Plan — decide which tool to use or which step to take.
4. Act — execute the action (search the web, send an email, call an API, etc.).
5. Repeat — examine the result and restart the loop until the job is done.

Loop in Action: BTC Price Email

Goal: "Find the current price of Bitcoin and email it to my boss."

```
goal    = "Find BTC price and email boss"
status = "incomplete"

while status != "complete":
    # 1. THINK
    thought = "I need to find the current price first."

    # 2. ACT
    price = tools.google_search("current bitcoin price")

    # 3. OBSERVE & RE-PLAN
    email = tools.contacts_db.get("Boss")

    # 4. ACT
    tools.email.send(to=email, body=f"BTC is {price}")
    status = "complete"
```

Without this loop the AI would guess using outdated training data. The loop lets the agent interact with the real world.

3. Goal-Oriented Reasoning

The agent's superpower is taking a vague, high-level objective and breaking it into logical sub-tasks — without a programmer hard-coding every step.

Example — The Travel Agent

User goal: "Book a trip to Tokyo for next Tuesday."

Agent's reasoning chain:

1. Check the user's calendar for availability.
2. Search for flights on Tuesday morning.
3. Check the user's budget in their preferences file.
4. If the flight is over budget, look for flights on Monday night instead.
5. Book via Stripe and send the confirmation via Gmail.

This if-then logic is generated dynamically by the LLM — not hard-coded.

4. Practical Examples

Example 1 — Research Agent (Correct Usage)

An agent is given a named tool (WeatherSearch) and asked a question. It Thinks, Acts (calls the tool), and Observes the result.

```
tools = [Tool(name="WeatherSearch", func=get_weather,
              description="Find current weather for a city.")]

agent = initialize_agent(tools, llm,
                        agent=AgentType.ZERO_SHOT_REACT_DESCRIPTION)

agent.run("Is it a good day for a picnic in San Francisco?")
```

Example 2 — The Infinite Loop (Common Mistake)

Bug vs. Fix
BUG: Agent has a 'Search Web' tool but no stopping condition — it keeps searching for the same thing because it cannot recognise a correct answer.
FIX: Add a clear Definition of Done to your system instructions AND set a max_iterations limit in code to prevent runaway API costs.

Key Takeaways

- LLMs are brains; Agents are workers. An agent uses an LLM to make decisions and tools to execute them.
- The loop is everything. Agents operate in a cycle of Observe → Think → Act.
- Tools are the hands. Without tools (Zapier, n8n, Python functions) an agent is just a chatbot.
- Reasoning is dynamic. Agents break down complex goals into smaller steps without needing every step hard-coded.
- Safety first. Always set limits on how many steps an agent can take to avoid infinite loops.

Reflection Questions

Use the space below to test your understanding:

1. In your own words, what separates an AI Agent from a plain LLM?

2. Draw or describe an Autonomous Loop for a task you care about.
3. What tool(s) would an agent need to complete your task in Question 2?